A FIELD PROJECT REPORT ON

AIR POLLUTION

Submitted

In partial fulfilment of the requirements for the award of the degree

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

By

T. Chiranjeevi	(241FA04329)
J. Deepak	(241FA04335)
N. Venu	(241FA04d90)
D.Kapil	(241FA04200)

Under the Guidance of

Mr. Kumar Devapogu

Assistant Professor, CSE

VIGNAN'S

FOUNDATION FOR SCIENCE, TECHNOLOGY & RESEARCH

(Deemed to be University) - Estd. u/s 3 of UGC Act 1956

Department of Computer Science and Engineering

School of Computing and Informatics

Vignan's Foundation for Science, Technology and Research (Deemed to be University) Vadlamudi, Guntur, Andhra Pradesh-522213, India.

April-2026

VIGNAN'S
FOUNDATION FOR SCIENCE, TECHNOLOGY & RESEARCH
(Deemed to be University) - Estd. u/s 3 of UGC Act 1956

CERTIFICATE

This is to certify that the field project entitled "**AIR POLLUTION**" is being submitted by **(T.Chiranjeevi & 241FA04329), (J.Deepak & 241FA04335), (N.Venu & 241FA04D90)**, and **(D.Kapil & 241FA04200)** in partial fulfilment of the requirements for the degree of Bachelor of Technology (B.Tech) in **Computer Science and Engineering** at Vignan's Foundation For Science Technology & Research (Deemed to be University), Vadlamudi, Guntur District, Andhra Pradesh, India.

This is a bonafide work carried out by the aforementioned students under my guidance and supervision.

Guide

Project Review Committee **Head Of the Department**

VIGNAN'S
FOUNDATION FOR SCIENCE, TECHNOLOGY & RESEARCH
(Deemed to be University) - Estd. u/s 3 of UGC Act 1956

DECLARATION

Date:

We hereby declare that the work presented in the field project titled "**AIR POLLUTION**", is the result of our own efforts and investigation. This project is being submitted under the supervision of **(Mr. Kumar Devapogu, Assistant professor, CSE)**, in partial fulfilment of the requirements for the Bachelor of Technology (B. Tech) degree in Computer Science and Engineering at Vignan's Foundation for Science, Technology and Research (Deemed to be University), Vadlamudi, Guntur, Andhra Pradesh.

T. Chiranjeevi	(241FA04329)
J. Deepak	(241FA04335)
N. Venu	(241FA04d90)
D. Kapil	(241FA04200)

TABLE OF CONTENTS

Chapter No.		Contents	Page No
1		Introduction	
	1.1	Problem Definition	
	1.2	Existing System	
	1.3	Proposed System	
	1.4	Literature Review	
2		System Requirements	
	2.1	Hardware & Software Requirements	
	2.2	Software Requirements Specification(SRS)	
3		System Design	
	3.1	Modules of System	
	3.2	UML Diagrams	
4		Implementation	
	4.1	Sample Code	
	4.2	Test Cases	
5		Results	
	5.1	Output Screens	
6		Conclusion	
7		References	

CHAPTER 1

Introduction

The **Air Pollution Awareness Website** project is designed to provide an online platform for users to learn about air pollution, its causes, effects, solutions, and the importance of **AQI (Air Quality Index)** in measuring air quality levels. This website helps people understand environmental issues, check pollution conditions, explore preventive measures, and spread awareness easily from anywhere. The project focuses on creating a user-friendly, attractive, and responsive website using web technologies such as HTML and CSS.

1.1 Problem Definition

The Air Pollution Awareness Website project aims to solve these problems by providing a centralized, user-friendly online platform where users can learn about air pollution, its causes, effects, solutions, and check AQI (Air Quality Index) levels easily. This website helps spread environmental awareness, improves access to pollution-related information, and encourages preventive actions, making learning about air quality faster, simpler, and more effective.

1.2 Existing System

- Air pollution information is mostly shared through newspapers, TV, or manual reports.
- People must visit government websites or offices to know pollution details.
- AQI updates are not always easy to access or understand.
- Limited awareness about causes, effects, and prevention methods.
- Information is scattered across different sources.
- Real-time pollution data is not available in one place.
- Many people do not regularly monitor air quality levels.
- Lack of 24/7 easy access to pollution awareness resources.
- Manual reporting methods can cause delays and errors.
- Public engagement and awareness remain low.

1.3 Proposed System

- A web-based Air Pollution Awareness Website accessible anytime and anywhere.
- Provides detailed and updated information about air pollution, AQI, causes, effects, and

solutions.

- Easy navigation with a user-friendly and attractive interface.
- Users can check air quality conditions and understand AQI levels online.
- Spreads awareness about preventive measures and environmental protection.
- Centralized platform for pollution-related information in one place.
- Reduces dependency on scattered or manual information sources.
- Improves accessibility, awareness, and learning efficiency.
- Responsive design that works on mobile, tablet, and desktop devices.
- Encourages people to take actions for a cleaner and healthier environment.

1.4 Literature Review

Various studies and existing environmental websites show that online awareness platforms play a crucial role in educating people about air pollution and its impact on health and the environment. Previous research highlights that web-based systems provide easy access to information about pollution sources, AQI (Air Quality Index), harmful effects, and preventive measures, which increases public awareness and understanding. Many existing air quality websites use responsive design, real-time data updates, and centralized information systems to improve accessibility and user engagement. These studies support the need for a user-friendly Air Pollution Awareness Website to meet the growing demand for digital environmental education and pollution monitoring services.

CHAPTER 2

SYSTEM REQUIREMENTS

2.1 Hardware & Software Requirements

Hardware Requirements

- iMac or Mac Book
- Processor: Mac M1 or higher
- RAM: Minimum 8 GB
- Hard Disk: Minimum 250 GB
- Internet Connection

Software Requirements

- Operating System: MAC OS 18 / MAC OS 26
- Web Browser: Google Chrome, Mozilla Firefox, or Safari
- Front-End Technologies: HTML, CSS
- Code Editor: Visual Studio Code / Terminal

2.2 Software Requirements Specification (SRS)

The SRS document outlines the functional and non-functional requirements of the system:

Functional Requirements (FR):

• Users can view information about air pollution, its causes, and effects.

• Users can check AQI (Air Quality Index) levels and understand air quality categories.

• Users can browse preventive measures and pollution control solutions.

• Users can access awareness content such as tips, charts, and environmental facts.

• Navigation links allow easy movement between sections (Home, AQI, Causes, Effects, Solutions, Contact).

• Users can submit feedback or contact details through an online form.

• Admin can update AQI data, pollution information, and awareness content (if backend implemented).

Non-Functional Requirements (NFR):

- **Usability:** The website should have a simple, attractive, and user-friendly interface.
- **Performance:** Pages should load quickly and work smoothly on different devices.
- **Security:** Contact form submissions and user data should be handled securely.
- **Availability:** The website should be accessible 24/7 via the internet.
- **Compatibility:** Should work on all major browsers (Chrome, Firefox, Edge).
- **Responsiveness:** The website should adapt properly to mobile, tablet, and desktop screens.

CHAPTER 3

SYSTEM DESIGNS

3.1 Modules of the System

1. **Home Module**
    a. Displays introduction about air pollution and website overview.
2. **AQI Module**
    a. Shows Air Quality Index levels and categories.
3. **Causes Module**
    a. Provides information about the main causes of air pollution.
4. **Effects Module**
    a. Explains the effects of air pollution on health and environment.
5. **Solutions Module**
    a. Displays preventive measures and solutions to reduce air pollution.

3.2 UML Design

Use Case Diagram

Actors: User, Admin

User Use Cases:

- View Home Page
- Check AQI
- View Causes
- View Effects
- View Solutions

CHAPTER 4

IMPLEMENTATION

4.1 Sample Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Manual AQI Checker</title>
<style>
body{
        font-family: Arial, sans-serif;
        background: linear-gradient(to right,#dff6ff,#ffffff);
        display:flex;
        justify-content:center;
        align-items:center;
        min-height:100vh;
        margin:0;
    }
    .container{
        background:white;
        width:90%;
        max-width:500px;
        padding:25px;
        border-radius:15px;
        box-shadow:0 5px 15px rgba(0,0,0,0.2);
        text-align:center;
    }
    h1{
        color:#0077b6;
    }
```

```
input{
    width:80%;
    padding:10px;
    margin:10px 0;
    border:1px solid #ccc;
    border-radius:8px;
}
}

button{
    padding:10px 18px;
    background:#0077b6;
    color:white;
    border:none;
    border-radius:8px;
    cursor:pointer;
    transition:0.3s;
}

button:hover{
    background:#023e8a;
    transform:scale(1.05);
}
.result{
    margin-top:20px;
    text-align:left;
    line-height:1.8;
}

.good{color:green; font-weight:bold;}
.moderate{color:orange; font-weight:bold;}
.poor{color:red; font-weight:bold;}
.verypoor{color:darkred; font-weight:bold;}
```

```
.circle{
   width:120px;
   height:120px;
   border-radius:50%;
   margin:20px auto;
   display:flex;
   justify-content:center;
   align-items:center;
   font-size:22px;
   font-weight:bold;
   color:white;
   animation:pop 1s ease;
}
green{background:green;}
.orange{background:orange;}
.red{background:red;}
.darkred{background:darkred;}
@keyframes pop{
   0%{transform:scale(0);}
   100%{transform:scale(1);}
}

@keyframes blink{
   0%,100%{opacity:1;}
   50%{opacity:0.5;}
}
.blink{
   animation:blink 1s infinite;
}
</style>
</head>

<body>
<div class="container">
```

```
    <h1>Manual AQI Checker</h1>
    <p>Enter city name</p>

    <input type="text" id="city" placeholder="Enter city name">
    <br>
    <button onclick="checkAQI()">Check AQI</button>

    <div class="result" id="output"></div>
</div>
<script>

function checkAQI(){

    let city = document.getElementById("city").value.toLowerCase().trim();
    let output = document.getElementById("output");

    if(city==""){
        output.innerHTML = "Please enter city name.";
        return;
    }
let data = {
        "delhi":   {aqi:320, status:"Very Poor", class:"verypoor", color:"darkred",
tip:"Stay indoors and wear mask."},
        "mumbai":  {aqi:180, status:"Moderate", class:"moderate", color:"orange",
tip:"Limit outdoor activities."},
        "hyderabad":{aqi:95, status:"Good", class:"good", color:"green", tip:"Air quality
is good. Enjoy outside."},
        "guntur":  {aqi:110, status:"Moderate", class:"moderate", color:"orange",
tip:"Sensitive people take care."},
        "chennai": {aqi:150, status:"Moderate", class:"moderate", color:"orange",
tip:"Reduce outdoor exercise."},
        "bangalore":{aqi:80, status:"Good", class:"good", color:"green", tip:"Fresh air.
Good for outdoor walk."},
```

```
        "kolkata": {aqi:220, status:"Poor", class:"poor", color:"red", tip:"Wear mask
outside."},
        "pune":    {aqi:100, status:"Moderate", class:"moderate", color:"orange",
tip:"Acceptable air quality."}
    };
let info;

    if(data[city]){
        info = data[city];
    }else{
        info = {aqi:120, status:"Moderate", class:"moderate", color:"orange",
tip:"Maintain clean environment and plant trees."};
    }

    output.innerHTML = `
        <div class="circle ${info.color} blink">${info.aqi}</div>
        <p><b>City:</b> ${city.toUpperCase()}</p>
        <p><b>Status:</b> <span class="${info.class}">${info.status}</span></p>
        <p><b>Suggestion:</b> ${info.tip}</p>
    `;
}

</script>

</body>
</html>
```

CHAPTER 5

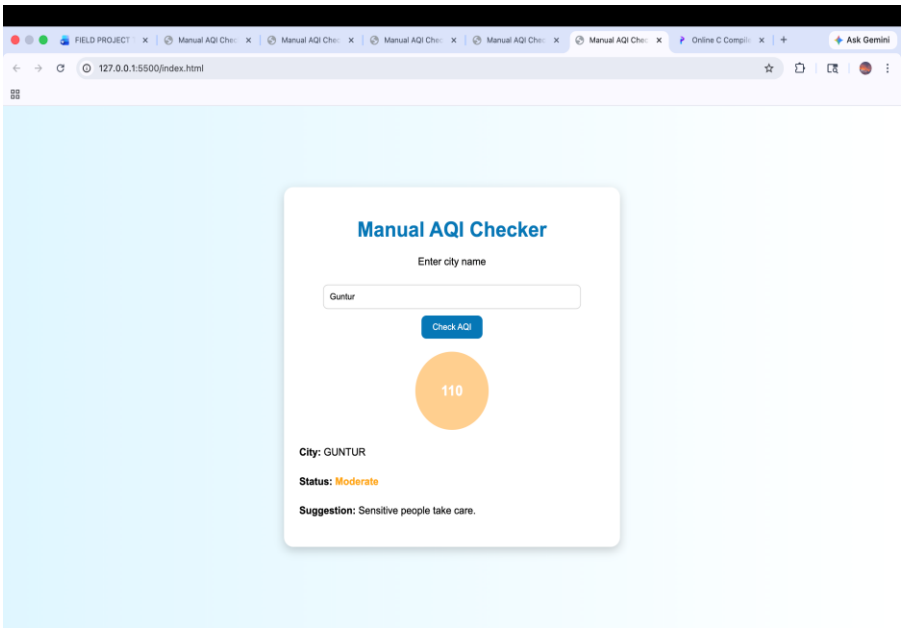
Result

5.1 <u>Output screen:</u>

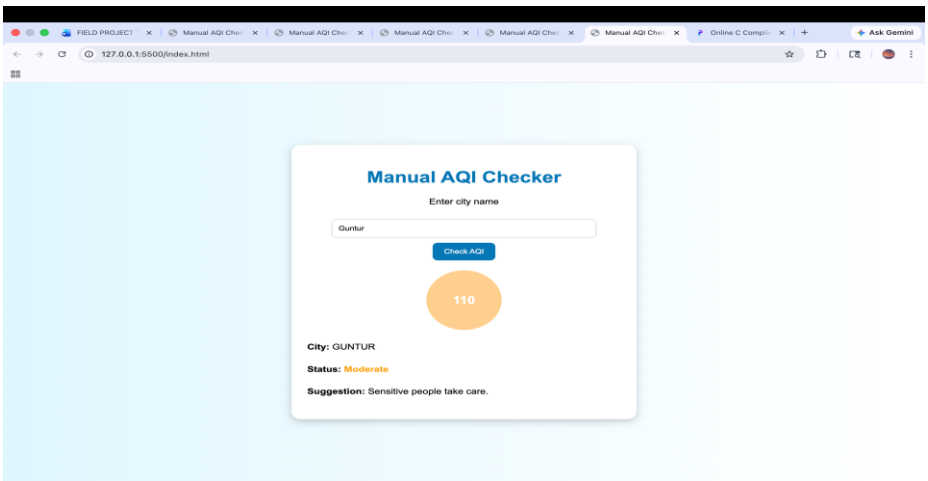
5.2 AQI is 320 in Delhi (Capital city of India)

5.3 AQI is 110 in Guntur (Chilli Capital of India)

5.2



5.3



CHAPTER 6
CONCLUSION

The Air Pollution Awareness Website project successfully provides a simple and user-friendly platform to educate people about air pollution, its causes, effects, solutions, and AQI levels. It helps users understand air quality conditions and encourages them to take preventive measures for a healthier environment. The project demonstrates how web technologies like HTML, CSS, and JavaScript can be used to create an attractive and responsive awareness system. Overall, this project contributes to spreading environmental awareness and promoting cleaner air for a better future.

CHAPTER 7

References

1. World Health Organization – Air Pollution and Health Reports.
2. United States Environmental Protection Agency – Air Quality Index (AQI) Guidelines.
3. Central Pollution Control Board – National Air Quality Monitoring Programme.
4. Open Weather – Air Pollution API Documentation.
5. United Nations Environment Programme – Global Environment and Pollution Studies.
6. HTML, CSS, and JavaScript official documentation and tutorials.